

Ex.no:4	EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION

Aim:

To perform data inspection, filtering, grouping, aggregation, descriptive statistical analysis, and correlation analysis using Pandas and visualize the results using Matplotlib.

Algorithm:

1. Import the required Pandas, NumPy and Matplotlib libraries.
2. Read the stud_data.csv file using Pandas.
3. Display the dataset and inspect the first few records.
4. Find the number of rows and columns and display the data types.
5. Generate the statistical summary of the dataset.
6. Filter the students based on marks and attendance conditions.
7. Sort the student records based on marks and attendance.
8. Group the data based on department and calculate average marks and attendance.
9. Calculate mean, median, minimum, maximum and standard deviation of marks.
10. Calculate the correlation between marks and attendance.
11. Create bar chart, line chart, scatter plot, histogram and box plot using Matplotlib.
12. Display the results and interpret the obtained data.

Program:

```
# Import libraries
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
# Read dataset
```

```
df = pd.read_csv("city_day.csv")
```

```
# Display dataset
```

```
print("Dataset:")
```

```
print(df)
```

```
# Display first five records
```

```
print("\nFirst Five Records:")

print(df.head())

# Display shape

print("\nShape of Dataset:")

print(df.shape)

# Display data types

print("\nData Types:")

print(df.dtypes)

# Display statistical summary

print("\nStatistical Summary:")

print(df.describe())

# Check missing values

print("\nMissing Values:")

print(df.isnull().sum())

# Filter records with PM2.5 greater than 100

print("\nRecords with PM2.5 greater than 100:")

print(df[df["PM2.5"] > 100])

# Filter records with PM10 greater than 100

print("\nRecords with PM10 greater than 100:")

print(df[df["PM10"] > 100])

# Sort records by PM2.5

print("\nRecords Sorted by PM2.5:")

print(df.sort_values("PM2.5", ascending=False).head(10))

# Sort records by PM10

print("\nRecords Sorted by PM10:")
```

```
print(df.sort_values("PM10", ascending=False).head(10))

# City-wise average PM2.5

print("\nCity-wise Average PM2.5:")

print(df.groupby("City")["PM2.5"].mean())

# City-wise average PM10

print("\nCity-wise Average PM10:")

print(df.groupby("City")["PM10"].mean())

# Calculate mean

print("\nMean PM2.5:", df["PM2.5"].mean())

# Calculate median

print("Median PM2.5:", df["PM2.5"].median())

# Calculate minimum

print("Minimum PM2.5:", df["PM2.5"].min())

# Calculate maximum

print("Maximum PM2.5:", df["PM2.5"].max())

# Calculate standard deviation

print("Standard Deviation of PM2.5:", df["PM2.5"].std())

# Display correlation matrix

print("\nCorrelation Matrix:")

print(df[["PM2.5", "PM10"]].corr())

# Display correlation between PM2.5 and PM10

print("\nCorrelation between PM2.5 and PM10:")

print(df["PM2.5"].corr(df["PM10"]))

# Bar chart - PM2.5 for first 10 records

plt.figure(figsize=(10, 5))
```

```
plt.bar(df["City"].head(10), df["PM2.5"].head(10))

plt.title("PM2.5 Levels")

plt.xlabel("City")

plt.ylabel("PM2.5")

plt.xticks(rotation=45)

plt.show()

# Line chart - PM10 for first 10 records

plt.figure(figsize=(10, 5))

plt.plot(df["City"].head(10), df["PM10"].head(10), marker="o")

plt.title("PM10 Levels")

plt.xlabel("City")

plt.ylabel("PM10")

plt.xticks(rotation=45)

plt.show()

# Scatter plot

plt.figure(figsize=(8, 5))

plt.scatter(df["PM2.5"], df["PM10"])

plt.title("PM2.5 vs PM10")

plt.xlabel("PM2.5")

plt.ylabel("PM10")

plt.show()

# Histogram

plt.figure(figsize=(8, 5))

plt.hist(df["PM2.5"].dropna(), bins=10)

plt.title("PM2.5 Distribution")
```

```
plt.xlabel("PM2.5")

plt.ylabel("Frequency")

plt.show()

# Box plot

plt.figure(figsize=(6, 5))

plt.boxplot(df["PM2.5"].dropna())

plt.title("PM2.5 Box Plot")

plt.ylabel("PM2.5")

plt.show()
```

Output:

```
Dataset:
   City      Date  PM2.5  PM10    NO    NO2    NOx    NH3  \
0  Ahmedabad  2015-01-01    NaN    NaN    0.92  18.22  17.15    NaN
1  Ahmedabad  2015-01-02    NaN    NaN    0.97  15.69  16.46    NaN
2  Ahmedabad  2015-01-03    NaN    NaN   17.40  19.30  29.70    NaN
3  Ahmedabad  2015-01-04    NaN    NaN    1.70  18.48  17.97    NaN
4  Ahmedabad  2015-01-05    NaN    NaN   22.10  21.42  37.76    NaN
...      ...      ...      ...      ...      ...      ...      ...
29526 Visakhapatnam  2020-06-27   15.02  50.94    7.68  25.06  19.54  12.47
29527 Visakhapatnam  2020-06-28   24.38  74.09    3.42  26.06  16.53  11.99
29528 Visakhapatnam  2020-06-29   22.91  65.73    3.45  29.53  18.33  10.71
29529 Visakhapatnam  2020-06-30   16.64  49.97    4.05  29.26  18.80  10.03
29530 Visakhapatnam  2020-07-01   15.00  66.00    0.40  26.85  14.05   5.20

   CO    SO2    O3  Benzene  Toluene  Xylene  AQI  AQI_Bucket
0   0.92  27.64  133.36    0.00    0.02    0.00   NaN          NaN
1   0.97  24.55  34.06    3.68    5.50    3.77   NaN          NaN
2  17.40  29.07  30.70    6.80   16.40    2.25   NaN          NaN
3   1.70  18.59  36.08    4.43   10.14    1.00   NaN          NaN
4  22.10  39.33  39.31    7.01   18.89    2.78   NaN          NaN
...      ...      ...      ...      ...      ...      ...      ...
29526  0.47   8.55  23.30    2.24   12.07    0.73  41.0          Good
29527  0.52  12.72  30.14    0.74    2.21    0.38  70.0  Satisfactory
29528  0.48   8.42  30.96    0.01    0.01    0.00  68.0  Satisfactory
29529  0.52   9.84  28.30    0.00    0.00    0.00  54.0  Satisfactory
29530  0.59   2.10  17.05    NaN     NaN     NaN  50.0          Good

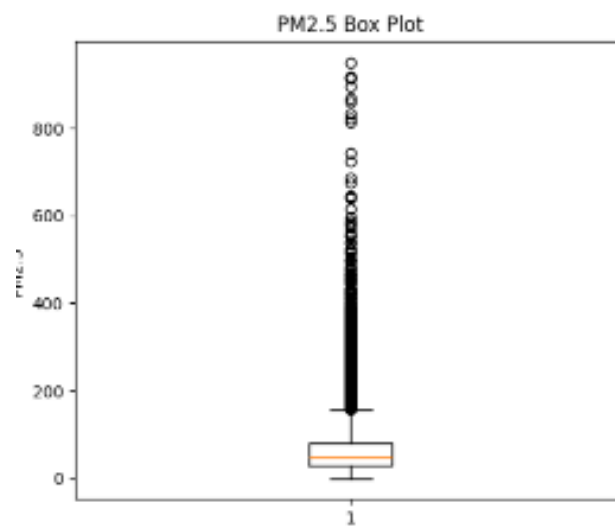
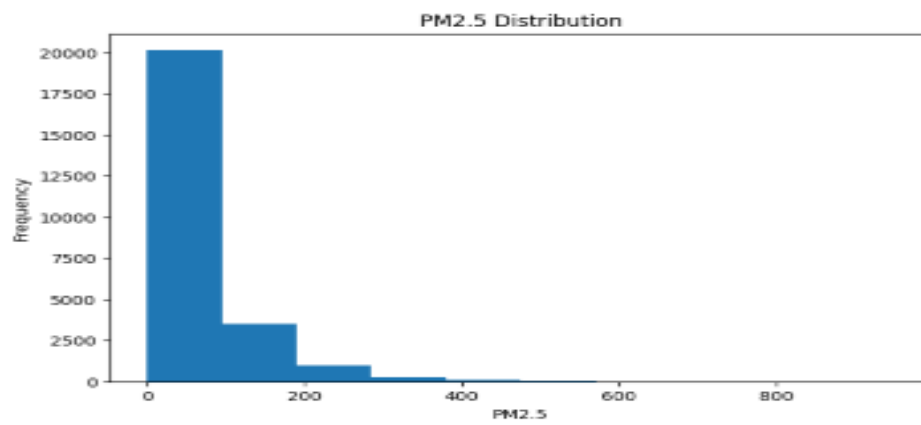
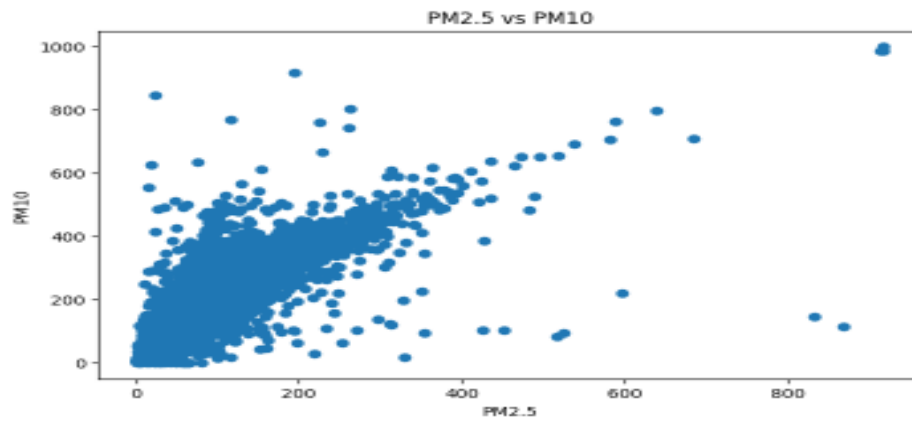
[29531 rows x 16 columns]

First Five Records:
   City      Date  PM2.5  PM10    NO    NO2    NOx    NH3    CO    SO2  \
0  Ahmedabad  2015-01-01    NaN    NaN    0.92  18.22  17.15    NaN    0.92  27.64
1  Ahmedabad  2015-01-02    NaN    NaN    0.97  15.69  16.46    NaN    0.97  24.55
2  Ahmedabad  2015-01-03    NaN    NaN   17.40  19.30  29.70    NaN   17.40  29.07
3  Ahmedabad  2015-01-04    NaN    NaN    1.70  18.48  17.97    NaN    1.70  18.59
4  Ahmedabad  2015-01-05    NaN    NaN   22.10  21.42  37.76    NaN   22.10  39.33

   O3  Benzene  Toluene  Xylene  AQI  AQI_Bucket
0  133.36    0.00    0.02    0.00   NaN          NaN
1   34.06    3.68    5.50    3.77   NaN          NaN
2   30.70    6.80   16.40    2.25   NaN          NaN
3   36.08    4.43   10.14    1.00   NaN          NaN
4   39.31    7.01   18.89    2.78   NaN          NaN

Shape of Dataset:
(29531, 16)

Data Types:
City      object
Date      object
PM2.5     float64
PM10      float64
NO        float64
NO2       float64
```



Result: